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Learning an Autonomous Drone Racing Policy via Differentiable Simulation

Master's Thesis

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Abstract

Todo

Keywords Unmanned Aerial Vehicles, Autonomous Drone Racing, Simulation to Reality Transfer, Reinforcement Learning, Domain Randomization

Abbreviations

DR Domain Randomization

Sim2Real Simulation-to-Reality

ADR Autonomous Drone Racing

OCP Optimal Control Problem

OCPs Optimal Control Problems

PMP Pontryagin's Maximum Principle

Contents

1 Introduction

1.1 Background

Give context about the problem

What is ADR?

- Navigate through a sequence of gates as fast as possible within the platform capabilities...

What are the challenges of ADR?

- **Where am I?** → Perception
  challenges: limited sensory data, noisy and uncertain partial observability
- **Where do I go?** → Planning
  safe efficient path planning
- **How do I get there?** →Control
  Handle the non-linear underactuated dynamics to execute the planned motion under actuator limits and model uncertainties

1.2 Related Work

Present the state of the art, how do the different research labs approach this problem

Open Questions

- How to improve sample efficency
- Generalization to multiple tasks (how to race while avoiding obstacles, how to race agains other drones)
- Transferability of the conterllers under unmodeled effects

1.3 Structure and Objectives

2 Preliminaries

This chapter covers important theoretical aspects on which this thesis is constructed. Section 2.1 formulates the OCP of drone racing and presents a theoretical solution based on the application of Pontryagin's Maximum Principle (PMP). The optimal solution derived from PMP serves primarily as an upper theoretical bound, as many of its underlying assumptions do not hold in real-world scenarios. Practical implementations must account for model uncertainties, disturbances, and system constraints that deviate from the idealized formulation.

In this context, Sections 2.3 and 2.4 review the state-of-the-art control approaches used to achieve high-performance drone racing, covering both classical control techniques and modern learning-based methods.

2.1 Optimal Control Problem

cite hurak, or well...some OC textbook

General Discrete-Time OCP

In general, a nonlinear discrete-time OCP can be formulated as the optimization of a running cost (or loss) function $L_k()$, plus a terminal cost function $\phi()$, subject to the system's dynamics $\mathbf{f}_k()$, and the set of permitted states $\mathcal{X}_k$, and controls $\mathcal{U}_k$.

$$\begin{aligned} \underset{\mathbf{X},\,\mathbf{U}}{\text{minimize}} \quad & \left(\phi(\boldsymbol{x}_N, N) + \sum_{k=0}^{N-1} L_k(\boldsymbol{x}_k, \boldsymbol{u}_k) \right) \\ \text{subject to} \quad & \boldsymbol{x}_{k+1} = \mathbf{f}_k(\boldsymbol{x}_k, \boldsymbol{u}_k), \quad k = 0, \ldots, N-1, \\ & \boldsymbol{u}_k \in \mathcal{U}_k, \quad k = 0, \ldots, N-1, \\ & \boldsymbol{x}_k \in \mathcal{X}_k, \quad k = 0, \ldots, N, \end{aligned} \tag{2.1}$$

Where

N is the final discrete time,
$\boldsymbol{x}_k \in \mathbb{R}^n$ is the state at the discrete time $k \in \mathbb{Z}$,
$\boldsymbol{u}_k \in \mathbb{R}^m$ is the control at the discrete time $k \in \mathbb{Z}$,
$\mathbf{X} = [\boldsymbol{x}_0, \ldots, \boldsymbol{x}_N]$ is the trajectory of states,
$\mathbf{U} = [\boldsymbol{u}_0, \ldots, \boldsymbol{u}_{N-1}]$ is the trajectory of control actions.

Minimum-Time OCP

This last formulation of an OCP can be further specified for the case of drone racing in several manners. Since the main objective is to complete a lap through the racing track as fast as possible, this problem can be written as minimum-time problem.

$$
\begin{aligned}
\underset{\mathbf{U},\ N}{\text{minimize}} \quad & \sum_{k=0}^{N-1} 1 \\
\text{subject to} \quad & \boldsymbol{x}_{k+1} = \mathbf{f}_k(\boldsymbol{x}_k, \boldsymbol{u}_k), \quad k = 0, \ldots, N-1, \\
& \boldsymbol{u}_k \in \mathcal{U}_k, \quad k = 0, \ldots, N-1, \\
& \boldsymbol{x}_k \in \mathcal{X}_k, \quad k = 0, \ldots, N,
\end{aligned}
\tag{2.2}
$$

The racing track restrictions, i.e. passing through a sequence of gates while avoiding obstacles, can be introduced as hard constraints under the set of permitted states $\mathcal{X}_k$ as

$$
\mathcal{X}_k = \left\{ \boldsymbol{x}_k \in \mathbb{R}^n \;\middle|\; \boldsymbol{p}_k \in \mathcal{G}_{g(k)}, \quad \boldsymbol{p}_k \notin \mathcal{O}_j, \quad \forall\, j \in \{1, \ldots, M\} \right\}, \tag{2.3}
$$

where

$\boldsymbol{p}_k \in \mathbb{R}^3$ is the position of the drone at time step k,
$\mathcal{G}_{g(k)}$ is the feasible region defined by gate g at time step k, with $g(k) \in \{1, \ldots, G\}$ denoting the index of the gate to be passed at k,
G is the total number of gates in the racing track,
$\mathcal{O}_j$ is the region occupied by the j-th obstacle,
M is the total number of obstacles.

Simplified Minimum-Time OCP

The anterior problem formulation (subsection 2.1.2), is rather complex. It has to solve simultaneously for an optimal trajectory that passes through all the gates and avoids obstacles. And, find the optimal control sequence achivable by the drone given its dynamics and actuator constrains. To simplify this problem, throughout this thesis, either an optimal trajectory or an optimal path will be assumed to be given by an external source. Thereafter, the Optimal Control Problems (OCPs) that we adress are

Trajectory Tracking OCP

$$
\begin{aligned}
\underset{\mathbf{U}}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \left\| \boldsymbol{x}_k - \boldsymbol{x}_k^{\text{ref}} \right\|_{\mathbf{Q}}^2 \\
\text{subject to} \quad & \boldsymbol{x}_{k+1} = \mathbf{f}_k(\boldsymbol{x}_k, \boldsymbol{u}_k), \quad k = 0, \ldots, N-1, \\
& \boldsymbol{u}_k \in \mathcal{U}_k, \quad k = 0, \ldots, N-1, \\
& \boldsymbol{x}_k \in \mathcal{X}_k, \quad k = 0, \ldots, N,
\end{aligned}
\tag{2.4}
$$

Path Progress OCP

2.2 Pontryagin's Maximum Principle

maybe solve it using a pmm and Pontryagin's Maximum Principle

Since it is impossible to perfectly model the state transition equation, and the state estimation is not perfect, the optimal solution breaks under real world conditions.

2.3 Classical Control approaches

Differential Flatness Based Control

Model Predictive Control

2.4 Learning-Based approaches

Model-Free Reinforcement Learning

Model-Based Reinforcement Learning

3 Methodology

This chapter presents the methodology used in this thesis to train the neural network policies. In section 3.1, we adress the core elements of the simulation architecture, define the quadrotor model, and explain how the Simulation-to-Reality (Sim2Real) gap is minimized by using Domain Randomization (DR). Furthermore, we introduce the implemented control abstraction levels. Section 3.2, outlines the training objectives used for Autonomous Drone Racing (ADR). Lastly, section 3.3 details how differentiability is leveraged to optimize the policies in order to minimize the training task's cost function.

3.1 Simulation Framework

Insert cool diagram

Figure 3.1: Your Caption

Quadrotor Model

Domain Randomization

Control Modes

Single Rotor Thrust

Collective Thrust and Body Rates

Linear Velocities and Yaw Rate

Adaptive Linear Velocities and Yaw Rate

3.2 Training Tasks

Trajectory Tracking

Path Progress

3.3 Differentiable Optimization

4 Results

4.1 Evaluation Criteria

4.2 Baseline Performance

4.3 Ablation Study

4.4 Transferability to MRS UAV System

4.5 Real World Validation

5 Conclusion

6 Future Directions

7 References

A Appendix A